Day 3 Mastery Quiz: Types, Casting, and Truthiness — Instructor Key

Text field → numeric cast → unchanged source → truthiness trap → safe evidence

Instructor key. Same layout as student copy; solutions filled in response boxes. Print duplex on one sheet.

Name		**Section**		**Date**	
Score	______/10		Mastery check	secure / developing / needs support	

A pump station sends intake values to the team as mixed Python values. Some numbers arrive as text, one approval field also arrives as text, and a blank comment may or may not mean anything. Your job is to decide what Python actually sees before tomorrow's boundary checks.

1 Reference code for Items 1–10

recognize · predict

```
# intake values from the station form
pump_id = "P-31"
target_text = "21.0"
reading_text = "19.8"
tolerance_text = "0.6"
approval_text = "False"
comment_text = ""
retry_count = 0
sensor_ready = True

# conversions and checks for the next day's limits
target_lpm = float(target_text)
reading_lpm = float(reading_text)
tolerance_lpm = float(tolerance_text)
gap_lpm = target_lpm - reading_lpm
approval_flag = bool(approval_text)
comment_flag = bool(comment_text)
retry_flag = bool(retry_count)
```

2 Items 1–4: type and cast evidence

recognize · trace

Pt.	Prompt	My response
1	What is the Python type of `reading_text` before conversion?	**Key:** `str`.
1	What value and type are stored in `target_lpm` after the cast?	**Key:** Value `21.0`, type `float`.
1	What expression converts `tolerance_text` into a numeric value without changing the original text variable?	**Key:** `float(tolerance_text)`.
1	After all three casts run, what value is still stored in `target_text`?	**Key:** The original string `"21.0"`.

3 Items 5–6: calculate only after safe conversion

predict · explain

Pt.	Prompt	My response
1	What value is stored in `gap_lpm`?	**Key:** `1.2`.
1	If a teammate used `target_text + tolerance_text`, what value would Python produce, and why is that not the numeric upper limit?	**Key:** It produces string concatenation `"21.00.6"`, not numeric `21.6`.

4 Items 7–8: truthiness traps

predict debug

Pt.	Prompt	My response
1	What value is stored in `approval_flag`, and why?	**Key:** `True`, because `"False"` is a non-empty string.
1	What values are stored in `comment_flag` and `retry_flag`, and what do they have in common?	**Key:** Both are `False`; empty string and numeric zero are falsey.

5 Item 9: debug an unsafe intake note

debug test

Note to check

The record is approved because `approval_flag` is `True`, and the upper limit can be made from `target_text + tolerance_text`.

Pt.	Prompt	My response
1	Name both mistakes in the note: one truthiness mistake and one type/casting mistake.	**Key:** Truthiness mistake: non-empty `"False"` makes `approval_flag` True but does not prove approval. Type mistake: adding text fields concatenates strings instead of computing a numeric upper limit.

6 Item 10: write a safer evidence check

implement transfer

Pt.	Prompt	My response
1	Write one Python expression that checks the approval text explicitly instead of relying on truthiness.	**Key:** `approval_text == "True"` or the course-approved label check.

Support route, not scored

My issue is mostly: setup / Python syntax / type names / casting / string concatenation / truthiness / confidence / time.
My next small action is: check `type(...)` / rerun one expression / compare empty and non-empty strings / ask a partner / ask instructor.